Week 11 — Multiple regression II: categories and interactions

Business Forecasting

My code (.R) Notes

Can a category improve the forecast?

Week 10 forecast the arrival delay at Mexico City airport, `flights$arr_delay`, from two numbers: the departure delay and the scheduled flight time.

`flights$carrier` adds who flies the plane. It has three categories:

`flights$carrier`	**Airlines in the category**
"Domestic"	the Mexican airlines: Aeroméxico, Volaris, Viva Aerobus, Magnicharters, Mas
"US"	the US airlines: United, American, Delta, FedEx
"Other foreign"	the other 20 foreign airlines: Avianca, Copa, Air Canada, LATAM, Iberia, Air France, Lufthansa…

The category is the airline's country, not the route: an Aeroméxico flight from Los Angeles is "Domestic".

A 0/1 regressor has two categories. `flights$carrier` has three; `flights$weekday` has seven. Can a regressor with several categories make the forecast of the arrival delay more accurate? A smaller miss means better plans for gates and ground crews.

1 · A regressor with several categories

One variable, three groups: what `lm()` does with it, and how to read the result.



Exercise 1 — Mean delay by airline group

Which group of airlines has the highest mean arrival delay? Which has the lowest? How many minutes apart are they?

1. Compute the mean of `flights$arr_delay` for each of the three values of `flights$carrier`.
2. Compare each mean with the Domestic mean.

Exercise | Start Over | Show Hint | Run Code

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Exercise 2 — The airline group in a regression

`lm()` accepts a text column as a regressor. What do its coefficients mean?

1. Regress `flights$arr_delay` on `flights$carrier`.
2. Compare the intercept with the Domestic mean of Exercise 1.
3. Compare the coefficient on `carrierUS` with the gap between the US mean and the Domestic mean.
4. Which category has no coefficient of its own?

Exercise | Start Over | Show Solution | Run Code

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From categories to dummies

Two categories, as in week 10: one 0/1 column, a dummy, is enough. `flights$intl` is 1 for a flight from abroad and 0 for a flight from Mexico, like `dayoff` for Ecobici.

`flights$carrier` has three categories, and one 0/1 column cannot tell three groups apart. So we need several dummies: one per category, 1 if the flight is in that category and 0 if not. `lm()` builds them automatically.

$$\text{domestic}_i = \begin{cases} 1 & \text{if carrier}_i = \text{“Domestic”} \\ 0 & \text{otherwise} \end{cases} \quad \text{other}_i = \begin{cases} 1 & \text{if carrier}_i = \text{“Other foreign”} \\ 0 & \text{otherwise} \end{cases} \quad \text{us}_i = \begin{cases} 1 & \text{if carrier}_i = \text{“US”} \\ 0 & \text{otherwise} \end{cases}$$

Flight	carrier	domestic	other	us
Aeroméxico from Cancún	"Domestic"	1	0	0
United from Houston	"US"	0	0	1
Avianca from Bogotá	"Other foreign"	0	1	0
Volaris from Tijuana	"Domestic"	1	0	0

- Each flight has exactly one 1 among the three dummies: it belongs to one category.
- Two categories needed one dummy, not two: “from Mexico” is simply `intl` = 0. In the same way, `summary()` in Exercise 2 printed two dummies, not three: `lm()` left `domestic` out.

Why one category must be left out

Each flight belongs to exactly one group, so the three dummies always add up to 1: $\text{domestic}_i + \text{other}_i + \text{us}_i = 1$.

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} \text{Aeroméxico from Cancún} \\ \text{United from Houston} \\ \text{Avianca from Bogotá} \\ \text{Volaris from Tijuana} \end{matrix}$$

$$\text{domestic} + \text{other} + \text{us} = \begin{bmatrix} 1+0+0 \\ 0+0+1 \\ 0+1+0 \\ 1+0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Four flights, all three dummies
Columns: the 1s for the intercept, then domestic, other, us.

Add the three dummy columns
In each row exactly one dummy is 1. The sum is the blue column, flight by flight.

The column of 1s for the intercept is already in $\mathbf{X}$. What happens if all three dummies go in with it?

The column of 1s is then the sum of three other columns: $\mathbf{X}$ is not full rank (week 10). R drops one category automatically to avoid this; the one dropped is the category with no coefficient in `summary()`. Try it in R

With m categories: $m-1$ dummies, so $m-1$ coefficients. The category left out is in the intercept.

What each coefficient means

The category left out is the reference category. R takes the first in alphabetical order, “Domestic”, so the regression has dummies for the other two. $\hat{y}$ is the predicted `arr_delay`:

$$\hat{y}_i = b_0 + b_1 \times \text{other}_i + b_2 \times \text{us}_i$$

$$\hat{y}_{\text{Domestic}} = b_0 + b_1 \times 0 + b_2 \times 0 = b_0$$

$$\hat{y}_{\text{Other foreign}} = b_0 + b_1 \times 1 + b_2 \times 0 = b_0 + b_1$$

$$\hat{y}_{\text{US}} = b_0 + b_1 \times 0 + b_2 \times 1 = b_0 + b_2$$

A Domestic flight: every dummy is 0
Only the intercept is left. b_0 is the mean delay of Domestic flights, the reference.

An Other foreign flight: other = 1
$b_1 = \hat{y}_{\text{Other foreign}} - \hat{y}_{\text{Domestic}}$: the gap in mean delay between Other foreign and Domestic flights.

A US flight: us = 1
$b_2 = \hat{y}_{\text{US}} - \hat{y}_{\text{Domestic}}$: the gap in mean delay between US and Domestic flights.

The intercept is the mean of the reference category. Each coefficient is the gap between its category and the reference.

The coefficients of Exercise 2

The fitted line of Exercise 2:

$$\begin{aligned}\hat{y}_i &= b_0 + b_1 \times \text{other}_i + b_2 \times \text{us}_i \\ \hat{y}_i &= -15.30 + 5.74 \times \text{other}_i + 7.22 \times \text{us}_i\end{aligned}$$

$$\hat{y}_{\text{Domestic}} = \underbrace{-15.30}_{b_0} + \underbrace{5.74}_{b_1} \times \underbrace{0}_{\text{other}} + \underbrace{7.22}_{b_2} \times \underbrace{0}_{\text{us}} = -15.30$$

$$\hat{y}_{\text{Other foreign}} = \underbrace{-15.30}_{b_0} + \underbrace{5.74}_{b_1} \times \underbrace{1}_{\text{other}} + \underbrace{7.22}_{b_2} \times \underbrace{0}_{\text{us}} = -9.56$$

$$\hat{y}_{\text{US}} = \underbrace{-15.30}_{b_0} + \underbrace{5.74}_{b_1} \times \underbrace{0}_{\text{other}} + \underbrace{7.22}_{b_2} \times \underbrace{1}_{\text{us}} = -8.08$$

A Domestic flight: every dummy is 0
Only the intercept is left. b_0 is the predicted delay of a Domestic flight: the Domestic mean.

An Other foreign flight: other = 1
b_1 is the gap to the reference: Other foreign flights land 5.74 minutes later than Domestic flights.

A US flight: us = 1
b_2 is the gap to the reference: US flights land 7.22 minutes later than Domestic flights.

What a coefficient of 0 means

$$\hat{y}_i = b_0 + b_1 \, \text{other}_i + b_2 \, \text{us}_i$$

Each coefficient compares one group with the reference. A coefficient of 0 means no gap.

$b_2 = 0$: $\hat{y}_{\text{US}} = b_0 + b_1 \times 0 + 0 \times 1 = b_0 = \hat{y}_{\text{Domestic}}$

US = Domestic
A US flight gets the same prediction as a Domestic flight: on average they land equally late.

$\hat{y}_{\text{US}} - \hat{y}_{\text{Other foreign}} = (b_0 + b_2) - (b_0 + b_1) = b_2 - b_1$

Only against the reference
$b_2 = 0$ says nothing about US against Other foreign. That gap is $b_2 - b_1$.

$b_1 = b_2 = 0$: $\hat{y}_{\text{Domestic}} = \hat{y}_{\text{Other foreign}} = \hat{y}_{\text{US}} = b_0$

The category does not matter
All three groups have the same mean, so the category does not help the forecast.

$b_0 = 0$: $\hat{y}_{\text{Domestic}} = 0$

The intercept
Domestic flights land on time on average. It says nothing about the gaps.

In data an estimate is never exactly 0: here $\hat{b}_1 = 5.74$ and $\hat{b}_2 = 7.22$. Next week: is an estimate far enough from 0 to say that the groups differ?

2 · Reading the coefficients of a category

The same reading three more times: six airlines, eight careers, thirteen store chains.

Exercise 3 — Every airline its own category

`flights6` keeps the flights of the six airlines with the most flights (91% of the flights). `flights6$airline` holds the airline's name.

1. Regress `flights6$arr_delay` on `flights6$airline`. How many categories are there? How many dummies? Which airline is the reference category?
2. What is the mean arrival delay of an Aeroméxico flight?
3. Read the coefficient on Volaris. What is the mean arrival delay of a Volaris flight?
4. Which lands later on average, Delta or United? By how many minutes?
5. Which airline lands latest on average? Which earliest?

Exercise | Start Over | Show Hint | Run Code

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Six airlines, five dummies

The output of Exercise 3: six categories, five dummies. The reference is Aeroméxico, first in alphabetical order.

Row of summary()	Estimate	Airline mean
(Intercept)	-20.0	-20.0
airlineAmerican Airlines	12.9	-7.1
airlineDelta Air Lines	7.8	-12.2
airlineUnited Airlines	13.7	-6.3
airlineViva Aerobus	16.9	-3.1
airlineVolaris	12.5	-7.5

What is the mean arrival delay of an Aeroméxico flight?

> The intercept
> -20.0 minutes: for Aeroméxico every dummy is 0.

Read the coefficient on Volaris.

> A gap to the reference
> A Volaris flight lands 12.5 minutes later than an Aeroméxico flight: $-20.0 + 12.5 = -7.5$ minutes.

Which lands later, Delta or United? By how many minutes?

> Two gaps to the same reference
> United, by $13.7 - 7.8 = 5.9$ minutes. The intercept cancels: $(\hat{b}_0 + 13.7) - (\hat{b}_0 + 7.8)$.

Which airline lands latest? Which earliest?

> The largest and the smallest coefficient
> Viva Aerobus lands latest, 16.9 minutes after Aeroméxico. Aeroméxico lands earliest: every coefficient is positive.

Which career pays most?

`careers` holds 10,541 people right after graduating, from the ENOE (INEGI), 2018 to 2022, in eight professional occupations. `careers$career` is the job the person does, not the degree.

`careers$career`	**Occupations in the category**
“Accounting and finance”	accountants, auditors, financial analysts, economists
“Business and marketing”	administrators, human resources, marketing, tourism
“Doctors”	general practitioners, medical specialists, dentists
“Engineering and architecture”	civil, industrial, mechanical, chemical and mining engineers; architects
“Law and social sciences”	lawyers, political scientists, sociologists, historians
“Nursing”	nurses and medical technicians
“Software and IT”	software developers, database and network administrators
“Teaching”	high school and university teachers

`careers$income` is monthly labour income, in pesos of 2022. Also: `hours` (worked last week), `female` (1 for women), `age`, `year`.

Source: INEGI, Encuesta Nacional de Ocupación y Empleo 2018–2022; workers with income between 0 and 100,000 pesos a month; incomes of 2018–2021 moved to 2022 pesos with INEGI’s consumer price index (INPC).

Which career has the highest mean income, and which the lowest? By how many pesos a month? For a student choosing a career, that gap is the price of the choice.

The regression we will run

Eight categories, so seven dummies. “Accounting and finance”, first in alphabetical order, is the reference category: it gets no dummy, and its mean is in β_0.

$$\begin{aligned}\text{income}_i = \; & \beta_0 + \beta_1 \, \text{business}_i + \beta_2 \, \text{doctors}_i + \beta_3 \, \text{engineering}_i + \beta_4 \, \text{law}_i \\ & + \beta_5 \, \text{nursing}_i + \beta_6 \, \text{software}_i + \beta_7 \, \text{teaching}_i + u_i\end{aligned}$$

career	business	doctors	engineering	law	nursing	software	teaching	income
Accounting and finance	0	0	0	0	0	0	0	16,184
Doctors	0	1	0	0	0	0	0	1,222
Software and IT	0	0	0	0	0	1	0	12,948

- An accountant has every dummy at 0: the prediction is β_0.
- A doctor has $\text{doctors} = 1$ and the rest 0: the prediction is $\beta_0 + \beta_2$.
- `lm(income ~ career, data = careers)` builds the seven dummies for you.

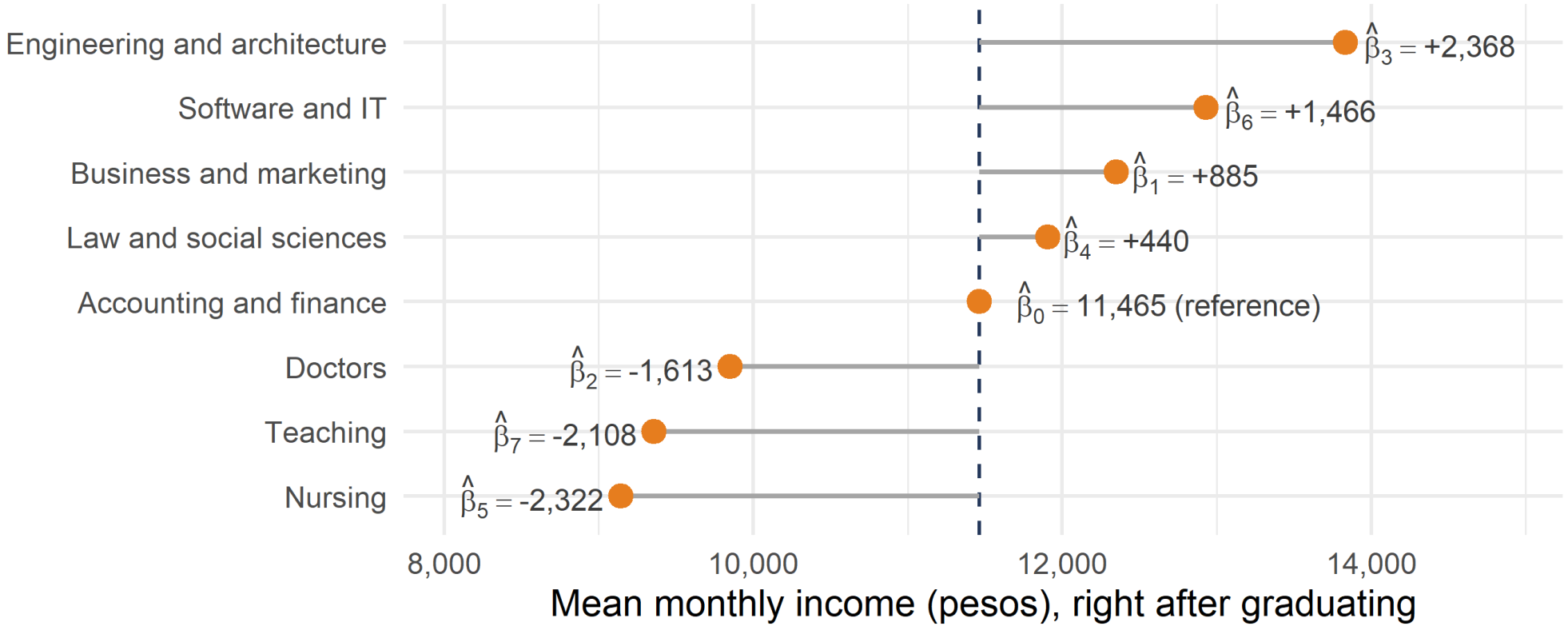
Exercise 4 — Which career pays most?

1. Regress `careers$income` on `careers$career`. Which category is the reference?
2. Which career has the highest mean income? The lowest? How many pesos above or below the reference?
3. Use the coefficients to find the mean income of a doctor.
4. How many pesos a month more does a doctor earn than a teacher, on average?

Exercise | Start Over | Show Hint | Run Code

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Pay by career



- Each coefficient is the distance to the dashed line: the gap to accounting and finance.
- Two careers that are not the reference: subtract their coefficients. A doctor against a teacher: -1,613 − (-2,108) = 495 pesos a month.
- Right after graduating, doctors earn less than accountants: many are still in social service or in training.

New data: shelf prices from PROFECO

PROFECO inspectors record shelf prices in stores all over Mexico ("Quién es Quién en los Precios", open data). `drinks` has 498 stores in 30 states, November 2025 to February 2026. One row is one item in one store in one month (the median of the visits).

Column	What it is
`price`	shelf price, pesos
`litres`, `units`	litres in the pack; containers in it (1, 6 or 12)
`product`, `brand`, `diet`	soda, water or beer; the brand; 1 if a sugar-free soda
`format`, `chain`, `store`	store type; chain; the store
`local`	1 for a store outside the national chains: public markets, regional supermarkets, Neto, Circle K
`rivals_2km`	other stores of the PROFECO survey within 2 km
`income`	municipal labour income per person, thousand pesos a month
`month`, `after`	the month; `after` is 1 from January 2026

Sources: PROFECO, Quién es Quién en los Precios; INEGI, Encuesta Intercensal 2025.

Which chain sells Coca-Cola cheapest?

`coke` keeps one product from `drinks`, the 355 ml can of Coca-Cola: 1,374 prices in 445 stores, November 2025 to February 2026. `coke$chain` is the chain of the store: 13 categories. “Other” groups the local stores outside the national chains.

Which chain has the lowest mean price for the can, and which the highest? By how many pesos? For a shopper, or for the bottler’s sales team, that gap is the price of the store.

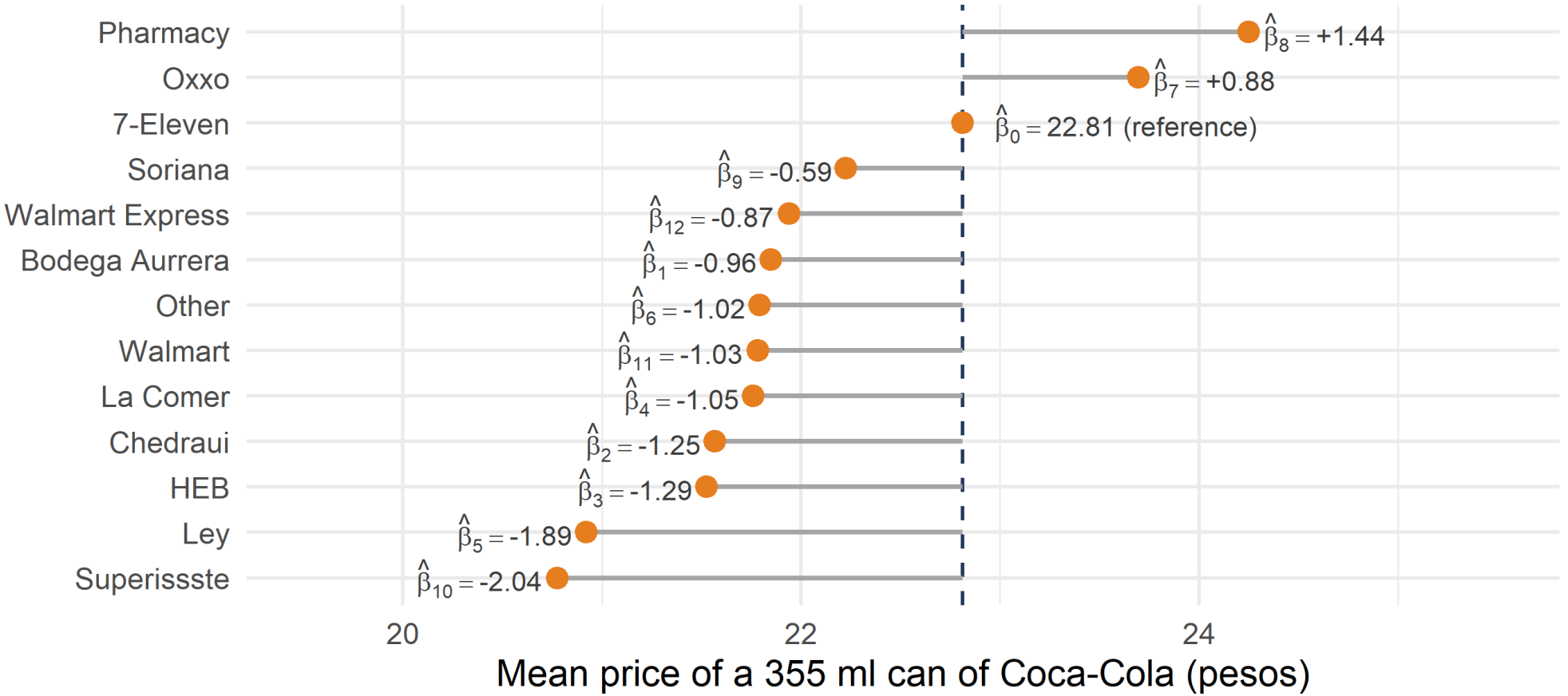
Exercise 5 — Where is a Coke cheapest?

1. Regress `coke$price` on `coke$chain`. How many categories are there? How many dummies? Which chain is the reference category?
2. Which chain has the lowest mean price? The highest? How many pesos from the reference?
3. Use the coefficients to find the mean price of a can at Oxxo.
4. How many pesos more is a can at Oxxo than at Walmart, on average?

Exercise | Start Over | Show Hint | Run Code

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Prices by chain



- Each coefficient is the distance to the dashed line: the gap to 7-Eleven, the reference.
- Pharmacies and Oxxo charge the most for the can; Superissste and Ley the least.

More on categories, in the appendix: Gaps with other regressors held fixed | Does a category improve the forecast?

3 · Interactions: one slope per group

So far a category moved the line up or down. Now it can also change the slope.

Does one more kilometre cost the same time?

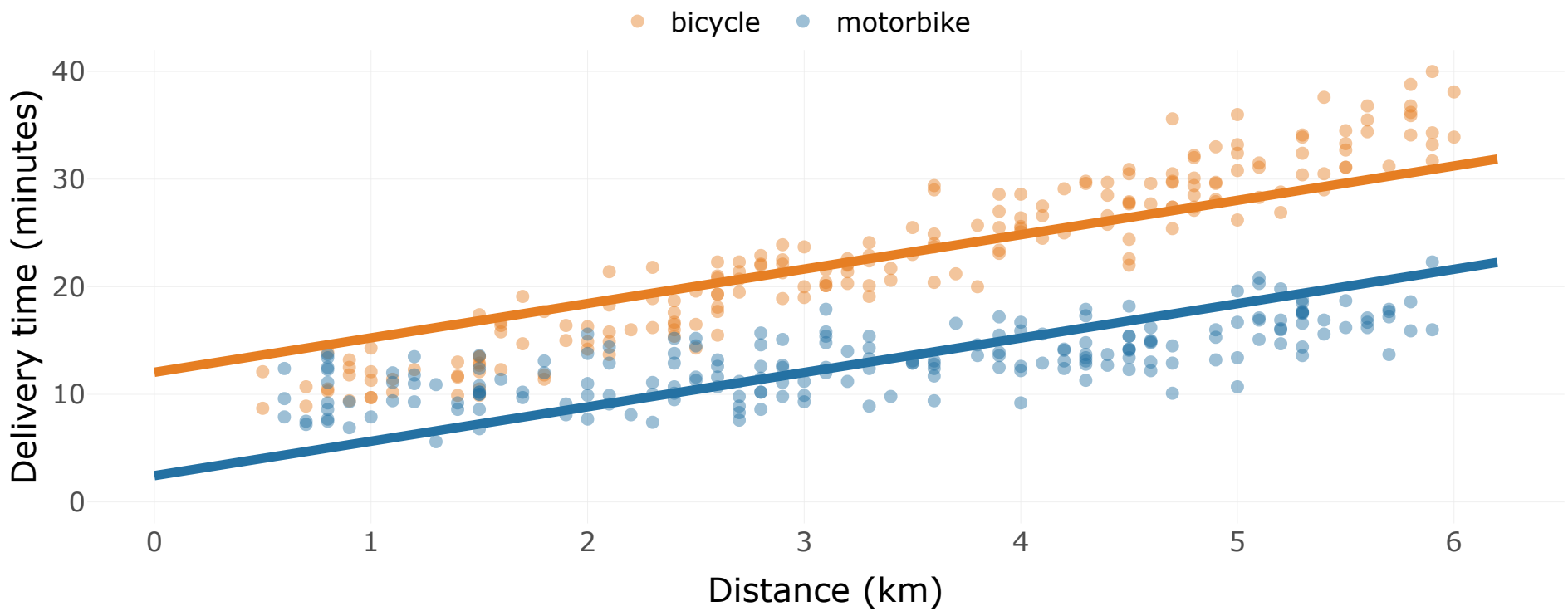
A delivery app promises each customer a delivery time. Its couriers ride bicycles or motorbikes. For each delivery the app records the distance and the minutes it took.

Does one more kilometre add the same number of minutes on a bicycle and on a motorbike? Would we expect the same effect?

No. A motorbike covers each kilometre faster, so each extra kilometre should add fewer minutes.

One slope: the same effect for both

The week 10 model, with a 0/1 regressor for motorbikes: $\text{minutes}_i = \beta_0 + \beta_1 \, \text{km}_i + \beta_2 \, \text{moto}_i + u_i$

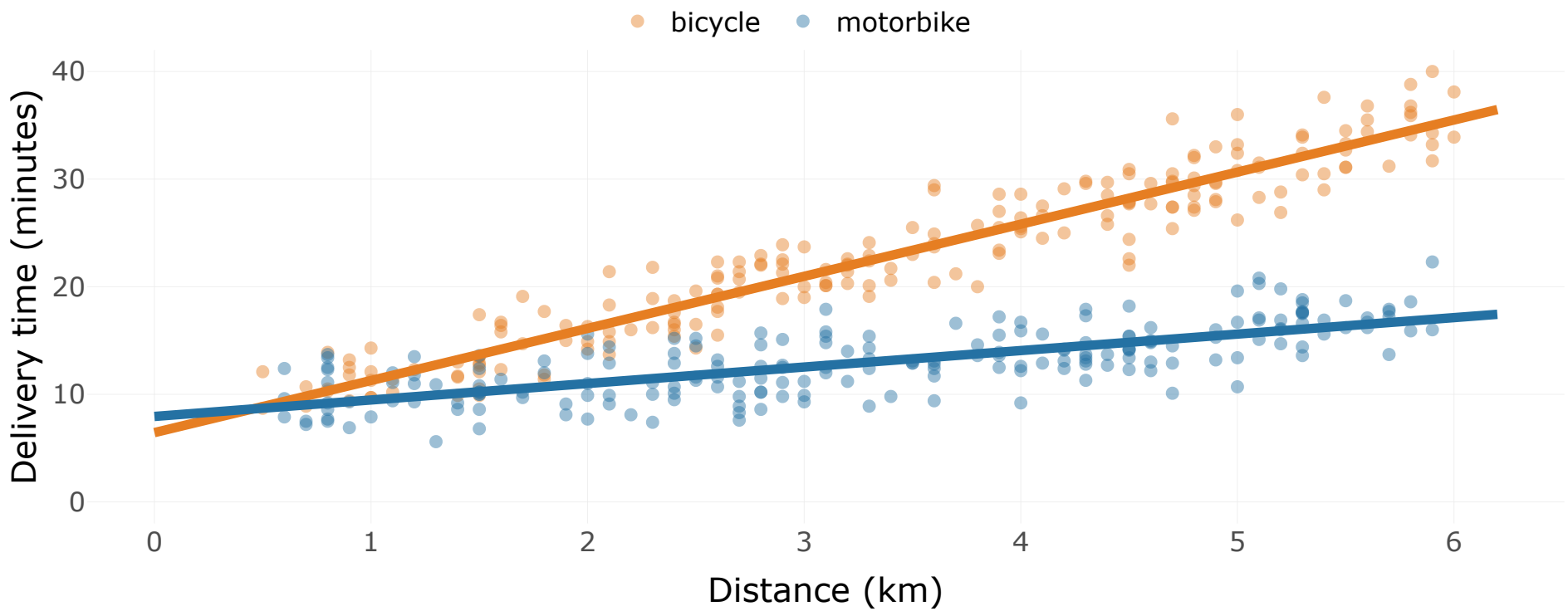


- One slope assumes one more kilometre adds the same β_1 minutes on both: here 3.20 minutes.
- The parallel lines miss both groups: too flat for the bicycles, too steep for the motorbikes.

Can we allow a different effect for each group? Yes: with an interaction.

Two slopes

The same deliveries, one line per group:



- Each group gets its own slope: the bicycle line is steeper.
- The two lines are not parallel, and they fit both groups.

How do we get two slopes in one regression?

One distance column, `km`, gives one slope for everyone. For two slopes we need two distance columns: `km` for everyone, and a second one that only applies to motorbikes. It equals the distance for a motorbike and 0 for a bicycle.

$$\mathbf{X} = \begin{bmatrix} 1 & 5.9 & 0 & 0 \\ 1 & 2.7 & 0 & 0 \\ 1 & 1.1 & 0 & 0 \\ 1 & 4.3 & 1 & 4.3 \\ 1 & 5.3 & 1 & 5.3 \\ 1 & 4.5 & 1 & 4.5 \end{bmatrix} \begin{matrix} \text{bicycle} \\ \text{bicycle} \\ \text{bicycle} \\ \text{motorbike} \\ \text{motorbike} \\ \text{motorbike} \end{matrix}$$

Columns: the 1s for the intercept, `km`, `moto`, and the new column: the distance for a motorbike, 0 for a bicycle.

The new column is $\text{km} \times \text{moto}$: the distance times the 0/1 regressor. The product of two regressors is called an interaction.

$$\text{minutes}_i = \beta_0 + \beta_1 \, \text{km}_i + \beta_2 \, \text{moto}_i + \beta_3 \, (\text{km}_i \times \text{moto}_i) + u_i$$

In R, two ways give the same coefficients:

- Build the column and add it: `deliveries$km_moto <- deliveries$km * deliveries$moto`, then `lm(minutes ~ km + moto + km_moto, data = deliveries)`.
- Or write the product in the formula: `lm(minutes ~ km * moto, data = deliveries)` adds `km`, `moto` and their product, which `summary()` calls `km:moto`.

What the regression gives for each group

$$\text{minutes}_i = \beta_0 + \beta_1\,\text{km}_i + \beta_2 \times \text{moto}_i + \beta_3 \times \text{km}_i \times \text{moto}_i + u_i$$

$$\begin{aligned}\text{minutes}_{\text{bicycle}} &= \beta_0 + \beta_1\,\text{km}_i + \beta_2 \times \quad 0 \quad + \beta_3 \times \text{km}_i \times \quad 0 \quad + u_i \\ &= \boxed{\beta_0 + \beta_1\,\text{km}_i} + u_i\end{aligned}$$

$$\begin{aligned}\text{minutes}_{\text{motorbike}} &= \beta_0 + \beta_1\,\text{km}_i + \beta_2 \times \quad 1 \quad + \beta_3 \times \text{km}_i \times \quad 1 \quad + u_i \\ &= \boxed{(\beta_0 + \beta_2) + (\beta_1 + \beta_3)\,\text{km}_i} + u_i\end{aligned}$$

$$\widehat{\text{minutes}}_{\text{bicycle}} = 6.44 + 4.84\,\text{km}_i$$

$$\widehat{\text{minutes}}_{\text{motorbike}} = (6.44 + 1.50) + (4.84 - 3.31)\,\text{km}_i$$

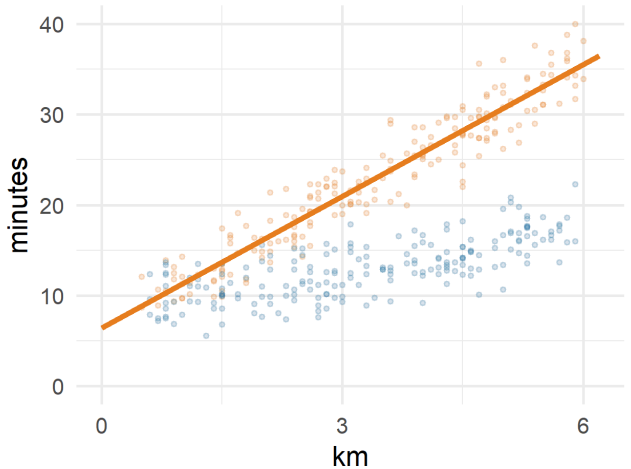
A bicycle: moto = 0
Both terms with moto drop out. The line starts at β_0 and rises β_1 minutes per kilometre.

A motorbike: moto = 1
β_2 moves the start of the line; β_3 changes its slope.

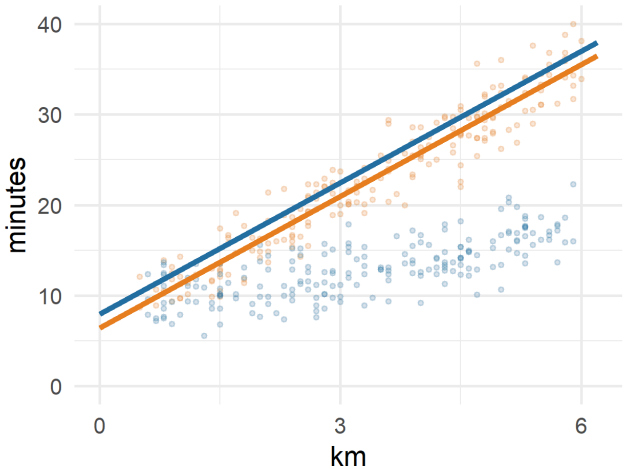
The OLS estimates
The motorbike line: $7.94 + 1.53\,\text{km}$.

What β_2 and β_3 do to the lines

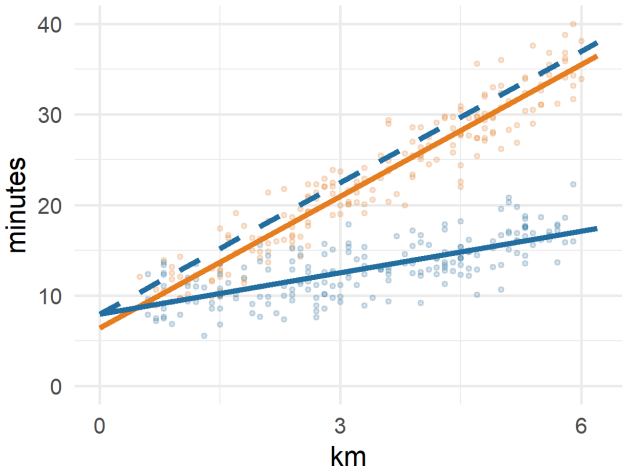
The 0/1 regressor works as a switch: anything multiplied by `moto` is 0 for bicycles and only counts for motorbikes.



$\beta_0 + \beta_1 \, \mathrm{km}$: one line, for bicycles



$+ \beta_2 \, \mathrm{moto}$: switched on for motorbikes, the line moves up



$+ \beta_3 \, (\mathrm{km} \times \mathrm{moto})$: an extra slope, switched on for motorbikes, turns the line

Exercise 6 — Two regressions or one?

`deliveries` holds 400 deliveries: `deliveries$minutes`, `deliveries$km`, and `deliveries$moto`, 1 for a motorbike and 0 for a bicycle.

1. Regress `minutes` on `km` using the bicycle deliveries only. Then do the same using the motorbike deliveries only.
2. Create `deliveries$km_moto`, the product of `deliveries$km` and `deliveries$moto`. Regress `deliveries$minutes` on `km`, `moto` and `km_moto`, using all deliveries.
3. From the coefficients of question 2, compute the start (at 0 km) and the minutes per km of each group. Compare them with the two regressions of question 1.

Exercise | Start Over | Show Hint | Run Code

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Two regressions in one

The output of Exercise 6, side by side:

	Separate regression	From the interaction
Bicycles: start	6.44	$\hat{\beta}_0 = 6.44$
Bicycles: minutes per km	4.84	$\hat{\beta}_1 = 4.84$
Motorbikes: start	7.94	$\hat{\beta}_0 + \hat{\beta}_2 = 6.44 + 1.50 = 7.94$
Motorbikes: minutes per km	1.53	$\hat{\beta}_1 + \hat{\beta}_3 = 4.84 - 3.31 = 1.53$

- The interaction gives exactly the two lines of the two separate regressions.
- So β_2 is the difference between the two starts, and β_3 the difference between the two slopes, motorbike minus bicycle.
- One regression instead of two: the difference in slopes now has its own coefficient, its own standard error and its own t-test.

Reading each coefficient

$$\widehat{\text{minutes}}_i = 6.44 + 4.84\,\text{km}_i + 1.50\,\text{moto}_i - 3.31\,(\text{km}_i \times \text{moto}_i)$$

- $\hat{\beta}_0 = 6.44$: a bicycle delivery of 0 km takes 6.4 minutes.
- $\hat{\beta}_1 = 4.84$: on a bicycle, each extra kilometre adds 4.84 minutes.
- $\hat{\beta}_2 = 1.50$: at 0 km, a motorbike delivery takes 1.50 minutes more than a bicycle one. It is the gap between the lines at 0 km only.
- $\hat{\beta}_3 = -3.31$: each extra kilometre adds 3.31 minutes less on a motorbike than on a bicycle, the reference group. It is a difference between two slopes, not the motorbike slope.

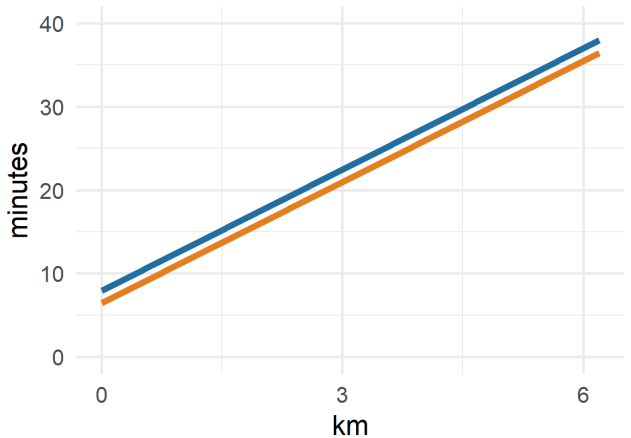
A negative $\hat{\beta}_3$ does not mean that a motorbike delivery gets shorter with distance. The motorbike slope is $\hat{\beta}_1 + \hat{\beta}_3 = 4.84 - 3.31 = 1.53$ minutes per kilometre: still positive, just smaller than on a bicycle.

The gap between the groups depends on the distance: $\hat{\beta}_2 + \hat{\beta}_3 \times \text{km}$. At 5 km it is $1.50 - 3.31 \times 5 = -15.06$ minutes: the motorbike is 15.1 minutes faster.

What a coefficient of 0 means in an interaction

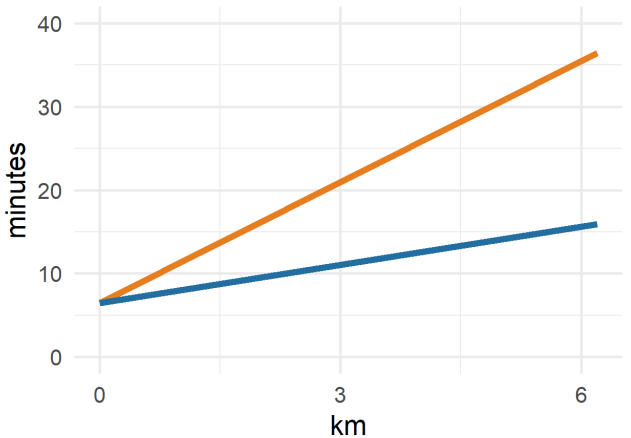
$$\text{minutes}_i = \beta_0 + \beta_1\,\text{km}_i + \beta_2\,\text{moto}_i + \beta_3\,(\text{km}_i \times \text{moto}_i) + u_i$$

The lines if one coefficient were 0, the others at their estimates:



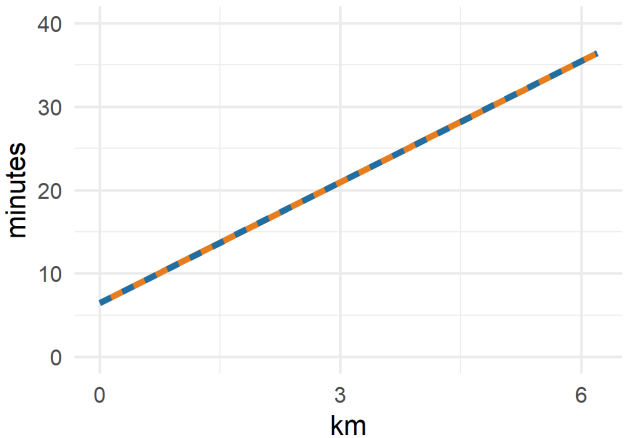
bicycle: $\beta_0 + \beta_1\,\text{km}$
motorbike: $(\beta_0 + \beta_2) + \beta_1\,\text{km}$

$\beta_3 = 0$: the same slope for both, parallel lines. One more kilometre costs the same time on either vehicle.



bicycle: $\beta_0 + \beta_1\,\text{km}$
motorbike: $\beta_0 + (\beta_1 + \beta_3)\,\text{km}$

$\beta_2 = 0$: the same start at 0 km. The lines open out from one point.



bicycle: $\beta_0 + \beta_1\,\text{km}$
motorbike: $\beta_0 + \beta_1\,\text{km}$

$\beta_2 = \beta_3 = 0$: one line for both. The vehicle does not matter.

$\beta_1 = 0$: distance does not matter for a bicycle, the reference group. The motorbike slope is then β_3 alone. Whether β_3 is 0 is the question of this section: the t-test on $\hat{\beta}_3$ answers it.

4 · Interactions in real data

The same model on real data: local competition, dating, and pay.

Does local competition matter?

`beer6` keeps one product from `drinks`, a six-pack of Dos Equis (six 355 ml bottles): 802 prices in 261 stores, November 2025 to February 2026.

- `beer6$rivals_2km` counts the other stores of the PROFECO survey within 2 km.
- `beer6$type` has three categories: “supermarket” (a supermarket chain, 238 stores), “convenience” (a convenience chain such as Oxxo or 7-Eleven, 12 stores) and “local” (a store outside the national chains, 11 stores). `beer6$local` is 1 for a local store.

Does one more store nearby change the price of the six-pack by the same number of pesos in all three types of store? Would we expect the same effect?

Probably not. A national chain, supermarket or convenience, sets its prices at head office, so the stores around one branch should barely matter. A local store sets its own price: alone, it can charge more; with rivals around, it must charge less.

The regression and the data

Three categories: two dummies, `conv` and `local`, each with its own product with `rivals`. Supermarkets are the reference.

$$\text{price}_i = \beta_0 + \beta_1\,\text{rivals}_i + \beta_2\,\text{conv}_i + \beta_3\,\text{local}_i + \beta_4\,(\text{rivals}_i \times \text{conv}_i) + \beta_5\,(\text{rivals}_i \times \text{local}_i) + u_i$$

type	rivals_2km	conv	local	rivals_conv	rivals_local	price
supermarket	0	0	0	0	0	112.0
supermarket	3	0	0	0	0	138.0
convenience	1	1	0	1	0	126.0
convenience	1	1	0	1	0	150.0
local	3	0	1	0	3	139.9
local	1	0	1	0	1	150.0

- Supermarket ($\text{conv} = \text{local} = 0$): $\beta_0 + \beta_1\,\text{rivals}$.
- Convenience ($\text{conv} = 1$): $(\beta_0 + \beta_2) + (\beta_1 + \beta_4)\,\text{rivals}$.
- Local ($\text{local} = 1$): $(\beta_0 + \beta_3) + (\beta_1 + \beta_5)\,\text{rivals}$.

🖊️ Exercise 7 — Competition in three types of store

1. Create `beer6$conv`, 1 for a convenience store (`beer6$type == "convenience"`). Then create `beer6$rivals_conv`, the product of `beer6$rivals_2km` and `beer6$conv`, and `beer6$rivals_local`, the product of `beer6$rivals_2km` and `beer6$local`.
2. Regress `beer6$price` on `rivals_2km`, `conv`, `local`, `rivals_conv` and `rivals_local`.
3. How many pesos does one more store nearby add to the price in a supermarket? In a convenience store? In a local store?
4. Which of the two slopes differs significantly from the supermarket slope at 5%?
5. With no other store within 2 km, how many pesos apart are a local store and a supermarket?

Exercise | Start Over | Show Hint | Run Code

```
1
```

What the stores nearby do to the price

- Supermarkets: -0.16 pesos per store nearby, not significant ($T = -1.38$). Head office sets the price.
- Convenience stores: -0.15 pesos per store nearby. $\hat{\beta}_4 = 0.02$ is close to 0 ($T = 0.04$): with $\beta_4 = 0$ they react like supermarkets, as chains should.
- Local stores: -3.10 pesos per store nearby ($T = -3.82$ for $\hat{\beta}_5$). Alone, a local store charges 3.06 pesos more than a supermarket; with rivals around, it must cut its price.
- The evidence comes from only 12 convenience and 11 local stores: most stores in the PROFECO survey are supermarkets.

Does attractiveness matter as much to everyone?

`dating` holds 544 ratings from 276 four-minute speed dates at Columbia University (Lock5Data: SpeedDating). After each date, each person rated from 1 to 10 how attractive they found their date, `dating$attractive`, and how much they liked them, `dating$like`. `dating$female` is 1 when the person rating is a woman.

Does one more point of attractiveness raise liking by the same amount when a man judges and when a woman judges?

$$\text{like}_i = \beta_0 + \beta_1 \, \text{attractive}_i + \beta_2 \, \text{female}_i + \beta_3 \, (\text{attractive}_i \times \text{female}_i) + u_i$$

attractive	female	attractive_female	like
6	0	0	6
7	0	0	8
5	1	5	5
8	1	8	8

A man judging: $\beta_0 + \beta_1 \, \text{attractive}$. A woman judging: $(\beta_0 + \beta_2) + (\beta_1 + \beta_3) \, \text{attractive}$.

Exercise 8 — Looks, by who is judging

1. Create `dating$attractive_female`, the product of `dating$attractive` and `dating$female`.
2. Regress `dating$like` on `dating$attractive`, `dating$female` and `dating$attractive_female`.
3. How much does one more point of attractiveness add to liking when a man judges? When a woman judges?
4. Is the difference significant at 5%?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Looks count for less when a woman judges

- A man judging: 0.71 points of liking per point of attractiveness. A woman judging: 0.59. With $\beta_3 = 0$ looks would count the same for both.
- $\hat{\beta}_2 = 0.79$ is the gap at attractiveness 0, a rating nobody can give. At 5 points the gap is $0.79 - 0.13 \times 5 = 0.15$.

Two categories: is the pay gap the same for doctors and nurses?

`health` holds the doctors and the nurses of all ages in the ENOE 2018–2022, income in pesos of 2022: 15,451 workers. Both regressors are 0/1: `health$female`, and `health$doctor` (1 for a doctor, 0 for a nurse).

Do women earn less than men by the same number of pesos among doctors and among nurses?

$$\text{income}_i = \beta_0 + \beta_1\, \text{female}_i + \beta_2\, \text{doctor}_i + \beta_3\, (\text{female}_i \times \text{doctor}_i) + u_i$$

Group	`female`	`doctor`	`female_doctor`	Predicted income
man, nurse	0	0	0	β_0
woman, nurse	1	0	0	$\beta_0 + \beta_1$
man, doctor	0	1	0	$\beta_0 + \beta_2$
woman, doctor	1	1	1	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

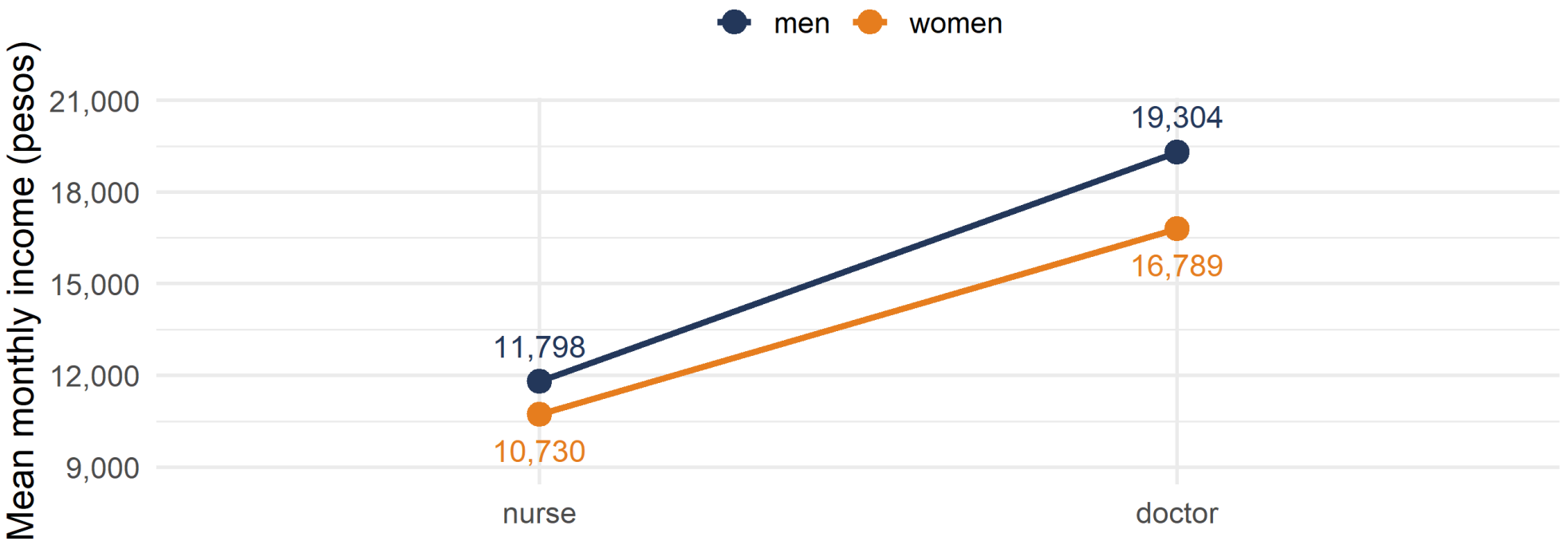
🖊️ Exercise 9 — The pay gap among doctors and nurses

1. Create `health$doctor`, 1 for `"Doctors"`, and `health$female_doctor`, the product of `health$female` and `health$doctor`.
2. Regress `health$income` on `health$female`, `health$doctor` and `health$female_doctor`.
3. Compute the four predicted incomes from the coefficients. Check one with `mean()`.
4. How large is the gender gap among nurses? Among doctors? Is the difference significant at 5%?

Exercise | Start Over | Show Solution | Run Code

```
1
```

The pay gap is wider among doctors



- Among nurses women earn 1,068 pesos a month less; among doctors 2,515 less. $\hat{\beta}_3$ is the difference. With $\beta_3 = 0$ the gap would be the same in both jobs, and the two lines parallel.
- Four groups, four coefficients: the regression reproduces the four means exactly.
- Two groups and two periods work the same way: the 2026 soda tax, in the appendix.

Two continuous regressors

You can also multiply two continuous regressors, for example `litres * income` in a price regression. The slope on litres then changes smoothly with income. We will not use this case in class.

More examples in the appendix:

Soda and water: a litre in each | Local stores and income | Two groups, two periods: the 2026 soda tax

What you should be able to do

- Explain how `lm()` turns a category into dummies, and why one category must be left out.
- Read the intercept as the mean of the reference category, and each coefficient as a gap to it.
- Explain why one slope assumes the same effect in every group, and build an interaction with arithmetic to give each group its own slope.
- Write the line of each group from an interaction, and read each coefficient: β_2 is the gap where the other regressor is 0, β_3 the difference in slopes.
- Read an interaction of two 0/1 regressors from the four group means.
- Say what a coefficient of 0 means: no gap to the reference category; in an interaction, $\beta_3 = 0$ is the same slope in both groups and $\beta_2 = 0$ the same start.

Next week: inference with several regressors. Is the gap between two careers significant? Do the seven career dummies matter together? How sure are we of a forecast?

Go to the exam practice →

Practice for this week

Weekly exercise set: `W11_exercises.pdf`, with solutions in `W11_solutions.pdf`.

You can already solve exercises 12 and 14, most of 8, 10 and 17, and 11 (b).

Not in these slides: 8 (f) and 9 use a 0/1 outcome, 9 (h) a squared regressor and 9 (i) standardized coefficients, which this course does not cover. 10 (d), 11 (a) and 16 compare two coefficients; 10 (h) and 11 (c) need intervals for a forecast; 11 (d)–(e), 13 and 15 test several coefficients at once or the assumptions: all week 12.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
8 (a)–(e)	Inflation forecasts, financial literacy and a college degree (`inflation_expectations`)	R with the data
10 (a)–(c), (e)–(g)	Weekly pay, children and marital status (`cps`)	R with the data
12	Reading an interaction of a 0/1 and a continuous regressor	By hand
17 (a)–(d)	Hours, pay and the gender gap (ENOE)	R with the data

Interactive versions: exam practice

Exam practice

Problems from this week's set, and a question from the Spring 2026 final. ← Back to the summary · ← Back to Practice for this week

Exam practice 1 — Who expects high inflation?

[P3 #8]

`inflation` holds 290 people in the United States, 2007–2008, from the `probstats4econ` package (`inflation_expectations`).

- `inflation$inflation_pred` — the inflation they expect next year, percentage points
- `inflation$age` — age, years
- `inflation$finlit_score` — financial literacy score, out of 12
- `inflation$collgrad` — 1 for a college graduate
- `inflation$famincome_hi` — 1 if family income is above 75,000 dollars

a. Regress `inflation_pred` on the four regressors. Which slopes are significant at 5%?

b. Interpret the slope on `finlit_score`.

c. Interpret the slope on `collgrad`.

d. Draw a histogram of the fitted values.

e. Add the product of `finlit_score` and `collgrad`. What is the p-value of the interaction? What is the slope on `finlit_score` for a college graduate, and for someone without a degree?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 2 — Pay, children and marriage

[P3 #10]

`cps` holds 4,013 people aged 30 to 59 from the 2019 Current Population Survey (`probstats4econ`). `cps$earnwk` is pay last week, dollars; `cps$gender` is “Female” or “Male”; `cps$ownchild` the number of own children at home; `cps$marstatus` has four categories; `cps$age` and `cps$educ` are in years.

a. Regress `earnwk` on `gender`. Interpret the two coefficients.

b. Add `ownchild`. What happens to the coefficient on `gender`?

c. Create `cps$female` (1 for women) and its product with `ownchild`. Regress `earnwk` on `female`, `ownchild` and the product. Interpret the coefficients.

d. Drop the product and add `marstatus`, `age` and `educ` to the model of (b). What happened to the `gender` coefficient?

e. How many dummies does `marstatus` use? Why?

f. Interpret the coefficient on the Married dummy.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 3 — Diet and exercise

[P3 #12]

100 people recorded their weight loss w (kg) after three months. $d = 1$ for those on a new diet; h is weekly hours of exercise. The fitted model is

$$\hat{w} = -1.4 + 7.2\,d + 1.5\,h + 3.4\,dh$$

Which statement is correct?

a. People who neither exercise nor follow the diet have a negative weight, which makes no sense.

b. People who follow the diet but do not exercise lose more than 5 kg.

c. For people not on the diet, one hour of exercise a week lowers weight by about 4.5 kg.

d. For people on the diet, one hour of exercise a week lowers weight by 10.7 kg.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 4 — Gym memberships

[P3 #14]

A gym has three memberships: limited, full and luxury. Muscle gain after three months is regressed on membership dummies and an intercept. The ANOVA table shows a total of 256 degrees of freedom and sums of squares of 500 (regression) and 1,200 (total).

What are the regression degrees of freedom and the sample size?

a. 2 and 257 (b) 3 and 257 (c) 4 and 256 (d) 2 and 254

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 5 — Hours, pay and the gender gap

[17]

`w12` holds the 2,000 ENOE workers of 2022 drawn in the set (Canvas, 02 Data, `ENOE_salaries.Rda`): `w12$Income` (monthly pesos), `w12$Hours_worked_lw` (hours last week) and `w12$Female` (1 for women).

a. Regress `Income` on `Hours_worked_lw` and `Female`. Report the three coefficients and R^2.

b. Interpret the coefficient on `Female`. What does this model assume about the slope on hours for the two groups?

c. Add the product of `Hours_worked_lw` and `Female`. Write the fitted line for men and for women.

d. Which group's pay rises faster with hours?

e. Is the difference in slopes significant at 5%? At 10%?

f. Write two sentences for an HR director.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 6 — A tariff on cars

[Spring 2026 final, Q13–Q18]

A Mexican manufacturer exports cars and boats to the United States. In the second quarter of 2025 the US put a tariff on cars, not on boats. `exports` holds the quarterly sales of each product, 2002 to 2026 (simulated exam data): `exports$Sales`, `exports$car` (1 for cars) and `exports$after_q2_2025` (1 from 2025 Q2).

a. Match each combination of (`car`, `after_q2_2025`) — (0,0), (1,0), (0,1), (1,1) — to boats or cars, before or after the tariff.

b. Build the product of `car` and `after_q2_2025` and regress `Sales` on the three regressors. What are the predicted boat sales before the tariff? After?

c. Interpret the coefficient on `after_q2_2025`.

d. What are the predicted car sales before the tariff?

e. Interpret the coefficient on `car`.

f. What is the effect of the tariff on car sales?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — optional

All three dummies in R; another reference category, by hand.

Appendix — category gaps with other regressors

The gap to the reference category when other regressors are held fixed.

Appendix — are the gaps fair?

Domestic flights land earliest. But the groups may differ in more than the airline: how late their flights leave, and how long they fly.

Which group lands latest for the same departure delay and the same scheduled time?

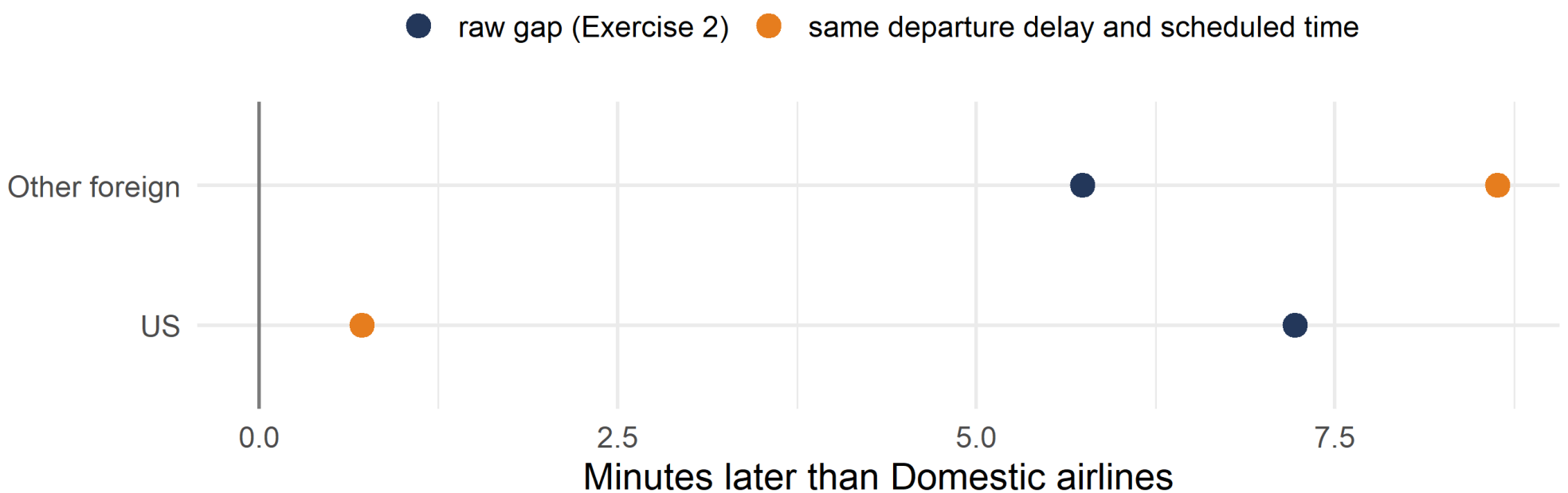
🖌️ Appendix exercise — Same departure, same schedule

1. Regress `flights$arr_delay` on `flights$dep_delay`, `flights$sched_time` and `flights$carrier`.
2. Compare each coefficient on `carrier` with Exercise 2. Which gap shrinks? Which grows?
3. Use the mean departure delay and the mean scheduled time of each group to explain why.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — raw gap and fair gap



- For the same departure delay and scheduled time, Other foreign airlines land 8.6 minutes later than Domestic airlines; US airlines only 0.7 minutes later.
- The rule of week 10 holds for each dummy: a gap moves when a left-out cause differs across the groups. The late departures pushed the US gap up; the long flights pushed the Other foreign gap down.
- Other foreign against US? $8.64 - 0.72 = 7.92$ minutes. Whether that gap is significant needs week 12.

Another reference category, by hand

← Back to the chains

Appendix — categories and fit

Back to the forecast. A category with many groups uses many degrees of freedom. Does it earn them?

Appendix — does the day of the week matter?

`flights$weekday` has seven categories; R takes Friday, first in alphabetical order, as the reference category. That is six dummies. `flights$carrier` needs two.

Which raises the adjusted R^2 of the week 10 forecast more: the day of the week or the airline group?

🖌️ Appendix exercise — Days, airlines and adjusted R^2

1. Regress `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`. Note the adjusted R^2.
2. Add `flights$weekday`. Note the adjusted R^2. Which days differ from Friday at 5%?
3. Replace `weekday` with `carrier`. Note the adjusted R^2.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — each group costs a degree of freedom

- A category with g groups adds $g - 1$ regressors. Adjusted R^2 charges for each one.
- The airline group is worth its two: the forecast's typical miss falls from 13.41 to 13.25 minutes.
- The day of the week is not: six dummies for a typical miss of 13.40 minutes.

Do the six days matter together, even if most are not significant one by one? That is one test of several coefficients at once: week 12.

← Back to the chains

Appendix exercise — All three dummies

1. Build the three 0/1 columns, for example `flights$us <- (flights$carrier == "US") * 1`.
2. Regress `flights$arr_delay` on two of them: `other` and `us`. Compare with Exercise 2.
3. Add the third, `domestic`. What does `summary()` print for it? Why?

Exercise | Start Over | Show Solution | Run Code

```
1
```

← Back to why one category is left out

Appendix — another reference category

To compare every group with the US airlines, leave US out instead of Domestic. Build the dummies with arithmetic (as in the appendix exercise) and choose which two go in:

```
reg <- lm(arr_delay ~ dep_delay + sched_time + domestic + other, data = flights)
summary(reg)
```

Dummy	Minutes later than US airlines
domestic	-0.72
other	7.92

- Every gap is now measured from the US airlines. Other foreign: 7.92, the same 8.64 − 0.72 as before.
- The fit, the slopes on `dep_delay` and `sched_time`, and every forecast are unchanged. Only the reference category moves.

← Back to the airlines

Appendix — is a litre worth the same in soda and in water?

`sw` holds the soft drinks and the bottled water of November 2025, one bottle at a time, up to 3 litres: 7,585 prices.

In week 10, a 0/1 regressor moved the whole line and kept its slope. Does an extra litre add the same number of pesos to a soda and to a bottle of water?

Appendix exercise — One slope or two?

1. Regress `price` on `litres` for the soda only, and then for the water only.
2. Create `sw$water`, 1 for water and 0 for soda. Regress `sw$price` on `sw$litres` and `sw$water`.
3. How many pesos per litre does each model give for soda? For water?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Appendix — two slopes for soda and water

$$\text{price}_i = \beta_0 + \beta_1\,\text{litres}_i + \beta_2\,\text{water}_i + \beta_3\,(\text{litres}_i \times \text{water}_i) + u_i$$

- For soda ($\text{water} = 0$): $\beta_0 + \beta_1\,\text{litres}$.
- For water ($\text{water} = 1$): $(\beta_0 + \beta_2) + (\beta_1 + \beta_3)\,\text{litres}$.
- The product of the two regressors is one more column in $\mathbf{X}$. Build it with arithmetic: `sw$litres_water <- sw$litres * sw$water`.

Appendix exercise — Build the interaction

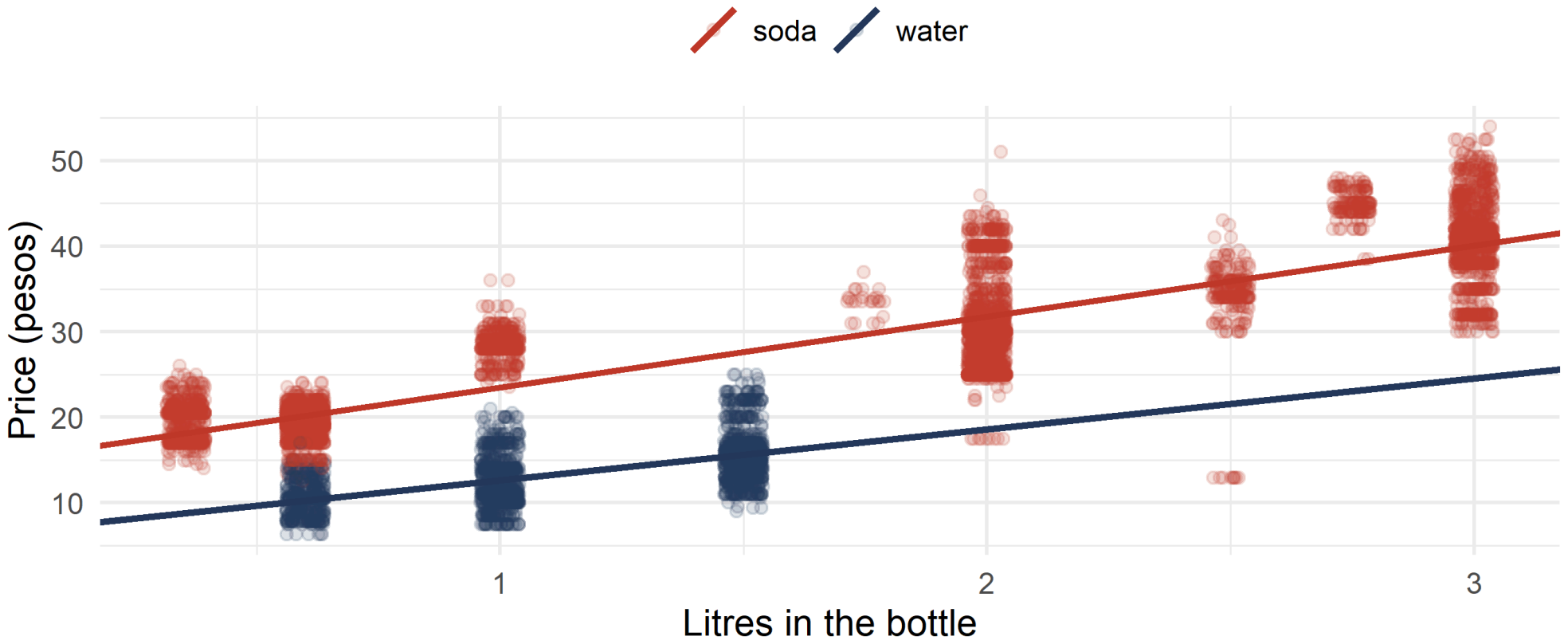
Does an extra litre add fewer pesos to water than to soda? By how many?

1. Create `sw$water` and `sw$litres_water`, the product of `sw$litres` and `sw$water`.
2. Regress `sw$price` on `sw$litres`, `sw$water` and `sw$litres_water`.
3. Recover the soda slope and the water slope of the previous exercise from the coefficients.
4. Is the difference between the two slopes significant at 5%?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — lines that are not parallel



- A 0/1 regressor alone moves the line. Its product with `litres` turns the line.
- The interaction answers “does the effect of size depend on the product?” Here, yes: soda charges more for each extra litre.
- The same product works for two continuous regressors: the slope of one then changes with the level of the other.

← Back to the interactions

Appendix — do local stores price to their neighbourhood?

Week 10 found that the national chains set national prices: municipal income barely moved the price of a soda.

`wat` holds the bottled water up to 1.5 litres, November 2025 to February 2026: 8,522 prices, 677 of them in local stores.

Does the price rise with municipal income in local stores, when it does not in the chains?

Appendix exercise — Local stores and income

1. Regress `wat$price` on `wat$litres` and `wat$income`.
2. Create `wat$income_local`, the product of `wat$income` and `wat$local`. Regress `wat$price` on `wat$litres`, `wat$income`, `wat$local` and `wat$income_local`.
3. How many pesos does 1,000 pesos more of municipal income add in a chain store? In a local store?
4. Above what income is a local store dearer than a chain store, for the same bottle?

Exercise | Start Over | Show Hint | Run Code

```
1
```

← Back to the interactions

Appendix — two groups, two periods

An interaction of two 0/1 regressors, and what a new tax did to prices.

Appendix — the 2026 tax on sugary drinks

- On 1 January 2026 the tax (IEPS) on sugary drinks rose from 1.6451 to 3.0818 pesos per litre.
- Sugar-free soft drinks, untaxed until then, now pay 1.5 pesos per litre. Water pays nothing.
- VAT of 16% is charged on top of the tax. In a 600 ml bottle the extra tax is $0.6 \times (3.0818 - 1.6451) \times 1.16 = 1.00$ peso.

Did stores pass the tax on to the shelf price? Comparing December with January is not enough: prices change in January for other reasons too.

Water pays no tax but goes through the same January. Its change shows what would have happened to soda without the tax.

🖊️ Appendix exercise — Did the price of a soda jump?

`tax600` holds the 600 ml bottles of soda and water in December 2025 and January 2026. `tax600$group` is “sugary soda”, “sugar-free soda” or “water”; `tax600$after` is 1 in January.

1. Compute the mean price of sugary soda and of water, in December and in January.
2. Keep the sugary soda and the water: `sw600 <- tax600[tax600$group != "sugar-free soda", ]`. Create `sw600$sugar` (1 for sugary soda) and `sw600$sugar_after`, the product of `sugar` and `after`.
3. Regress `sw600$price` on `sugar`, `after` and `sugar_after`. Match each coefficient to the four means.
4. Compare the coefficient on `sugar_after` with the extra tax in the bottle, 1.00 peso.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — the difference in differences

600 ml bottle	December	January	Change
sugary soda	19.32	21.04	1.71
water	10.63	10.93	0.29
difference	8.69	10.11	1.42

$$\text{price}_i = \beta_0 + \beta_1 \, \text{sugar}_i + \beta_2 \, \text{after}_i + \beta_3 \, (\text{sugar}_i \times \text{after}_i) + u_i$$

- Four groups, four coefficients: the regression reproduces the four means exactly.
- β_3 is a difference in differences: (soda's change) − (water's change).

Appendix exercise — Sugar-free soda

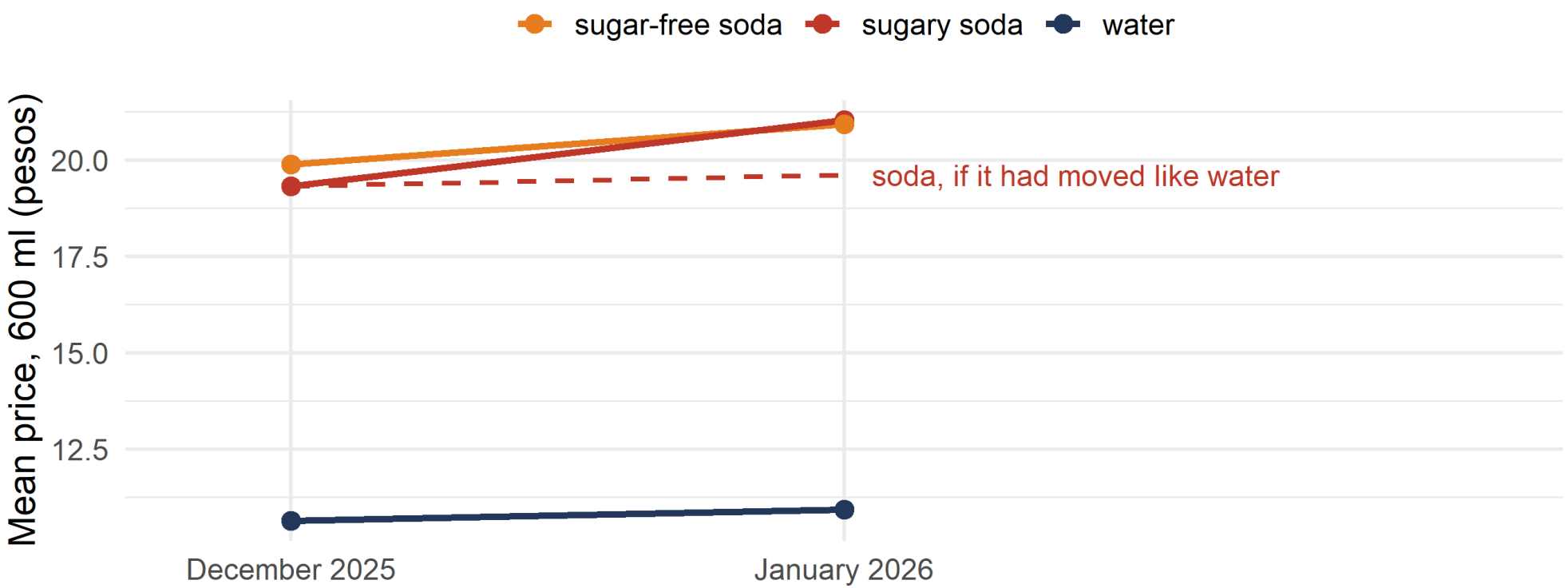
Sugar-free soda went from no tax to 1.5 pesos per litre: 1.04 peso in a 600 ml bottle, with VAT. Did the stores pass that on?

1. Keep the sugar-free soda and the water: `dw600 <- tax600[tax600$group != "sugary soda", ]`.
2. Build the 0/1 column for sugar-free soda and its product with `after`.
3. Regress the price on the three regressors and read the extra change for sugar-free soda.
4. Compare it with 1.04 peso, and with the sugary soda of the previous exercise.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — what the tax did



- The dashed line is the comparison: soda moving as water did. The gap at January is $\beta_3 = 1.42$ pesos.
- For a bottler or a retailer: the 2026 tax reached the shelf in full, and more, on sugary drinks.
- It only works if soda and water would have moved together without the tax. Water is the stand-in for what soda would have done.

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